Supporting Results
Context-Aware Supervision for Selective Sensor Reconstruction

6 November 2026

This document retains detailed results behind the manuscript's locators E1–E4 and M1. The evidence comes from closed predictions and evaluator records. The exact method, source identities and reproduction code accompany this document on GitHub[1]. The section locators refer to topics shared with that material, while tables and figures in this document use the distinct prefix S.

E1. History calibration and acceptance thresholds

The auxiliary ladder compares four normalizers using the same ridge learner, normal-reference data, penalty, grouped folds and scale bounds. Its shared-history ridge fit is a separately identified explanatory fit, not the shared/shrunk normalizer of *Reference*. The AE, scaler, TTO settings, candidate ranking, context checks and numerical-validity predicates remain fixed. The normalizers and their transforms are fitted on the retained normal-reference data.

The risk-refitted comparison uses each level's retained development-selected gap. The fixed-gap comparison instead uses the full-*Context* gap at every level. In both comparisons, normalization changes the event score, its calibrated limit and event admission. The ladder therefore does not isolate a causal history effect with every downstream boundary held constant.

Table S1: The complete ladder on all 193,350 primary faults per row. Here B counts correct-beneficial proposals, H counts harmful proposals and G is mean signed selected-channel benefit in percent of training range. The displayed benefits are rounded to two decimals.

Acceptance gap	History features	B	H	G
Risk-refitted	Intercept only	52,165	6,427	3.73
	Shared history	54,917	8,429	3.64
	Shared + component state	57,315	9,804	3.55
	Full component history	57,966	8,653	3.70
Fixed full-*Context*	Intercept only	52,325	6,490	3.74
	Shared history	54,868	6,886	3.76
	Shared + component state	57,009	7,632	3.74
	Full component history	57,966	8,653	3.70

With the full gap fixed, adding history beyond current component state adds 957 beneficial and 1,021 harmful proposals. With the level-specific gaps, the same final step adds 651 beneficial proposals and removes 1,151 harms. The change in the harm comparison follows the accepted sets produced by threshold selection and event admission. The full-history fit has more beneficial proposals than the intercept fit, but greater harm and lower mean signed benefit in both comparisons.

[1] `https://github.com/B-Ludwig/Context-Aware_Supervision_for_Selective_Sensor_Reconstruction`

E2. Configuration, checkpoint and error detail

The primary assessment contains 193,350 synthetic fault observations per policy across twelve configurations and five checkpoints. The configurations share 132 source periods from sixteen files or batches. The observations reuse channels, fault variants, source periods and checkpoints. The summaries below are descriptive comparisons of these dependent observations.

Table S2: The complete configuration breakdown for *Context*. The rates use all N primary faults. The count changes ΔB and ΔH compare *Context* with *Reference*, so positive ΔH means more harm. The HAI configurations are DS1–DS10.

Unit	N	D/N (%)	B/N (%)	ΔB	ΔH
DS1	8,250	93.55	87.16	+152	+199
DS2	12,000	84.57	79.24	+18	+180
DS3	6,000	93.35	71.73	+3	-2
DS4	6,000	91.37	63.68	+10	+13
DS5	15,675	81.03	16.91	+363	+232
DS6	28,800	19.70	7.68	-67	+39
DS7	13,200	78.74	23.78	+1,395	+113
DS8	22,275	83.83	18.93	+3,147	+1,704
DS9	21,600	76.87	43.84	+387	-486
DS10	37,950	58.48	6.11	-980	-2,484
FourTank	11,700	79.84	51.63	-49	-8
SWaT	9,900	43.54	31.24	-3	-1

The additional beneficial proposals are concentrated in DS7 and DS8, which add 1,395 and 3,147 respectively. The remaining configurations collectively lose 166 beneficial proposals. The DS8 gain includes 1,704 additional harms. In DS10, *Context* loses 980 beneficial proposals but removes 2,484 harms. The reduction in DS10 damage drives the incremental pooled signed benefit, although DS10 remains negative under both policies.

Table S3: The paired *Context* minus *Reference* changes at each checkpoint. Each row contains 38,670 primary faults. The change ΔG is in percentage points of training range per case.

Checkpoint	ΔB	ΔH	ΔG
23	+824	+233	+0.10
42	+724	−782	+0.12
101	+638	−1,037	+0.84
2027	+1,346	+495	−0.25
31415	+844	+590	−0.06

The coverage gain spans 1.65–3.48 percentage points across the checkpoints. The signed benefit improves at three checkpoints, while harm decreases at two. For DS6, *Context*'s beneficial coverage ranges from 1.18% to 12.90%. Among the 132 source-period groups, beneficial counts increase, tie and decrease in 72, 20 and 40 groups respectively. These groups can share a recording or batch and do not constitute independent plant replications.

The pooled fault mixture contains 80% drift cases. For *Context*, pooled beneficial coverage is 29.98%, compared with *Reference*'s 27.72%. When configurations receive equal weight, the corresponding coverages are 41.83% and 39.52%. The configuration means of within-configuration target precision are 87.83% and 88.91%, while the means of within-configuration harm proportions are 13.57% and 12.69%. The pooled mean signed benefits are 3.70% for *Context*, 3.55% for *Reference* and 5.03% for *Residual dominance*. Giving each configuration equal weight re-

verses the last comparison, yielding 6.14% for *Context* and 5.33% for *Residual dominance*. These choices describe weighting sensitivity rather than deployment prevalence or industrial error costs.

For the family-balanced summary, each of drift, exact stuck, noisy stuck and quantized stuck receives equal weight after within-family pooling. The resulting mean beneficial coverage is 19.06%. The reported 25.32% harm summary is the arithmetic mean of the four within-family conditional harm rates $H_f/N_{P,f}$. It is not the conditional harm rate obtained by dividing globally reweighted harm and proposal totals. The precise within-group aggregation must therefore accompany the value.

The signed benefit is selected-channel MAE before minus after reconstruction, divided by that channel's training range and averaged over all primary cases. An abstention contributes zero. A harmful wrong-target reconstruction remains negative, while missing clean-reference quality remains unavailable. All primary proposals have measured quality. Of *Context*'s 8,653 harmful primary proposals, 7,684 select the wrong channel. Among harmful FourTank proposals, median degradation is 54.21% of the selected channel's training range. The remaining injected-target error is measured separately from this additional wrong-channel damage. Across the assessment, applying proposals offline reduces median injected-target error from 5% to 2% of training range but leaves its 95th percentile at 50%. With the target supplied by the evaluator, TTO improves 79.32% of all cases. This known-target result does not establish the coverage achieved by the supervisor.

Figure S1 shows detection and useful reconstruction coverage across drift magnitudes. The severity diagnostics also separate detection from reconstruction. For DS8, event counts rise from 11,757 under *Reference* to 18,673 under *Context* out of 22,275 faults, while harms rise from 188 to 1,892. Across HAI configurations, mean beneficial coverage falls from 52.44% at 2% drift to 40.40% at 50% drift despite higher detection. The fraction of otherwise qualifying candidates withheld by the process flag rises from 4.48% to 50.39% at those magnitudes. For exact stuck, 6,080 of 12,890 cases introduce no change in the evaluation window and remain in the denominator. In SWaT, 81 of 570 changed exact-stuck cases are detected, compared with 85 of 660 overall. With the target known, TTO improves 234 of those 660 cases. The retained outcomes do not distinguish the contributions of thresholding, input copying and weak redundancy to these missed detections.

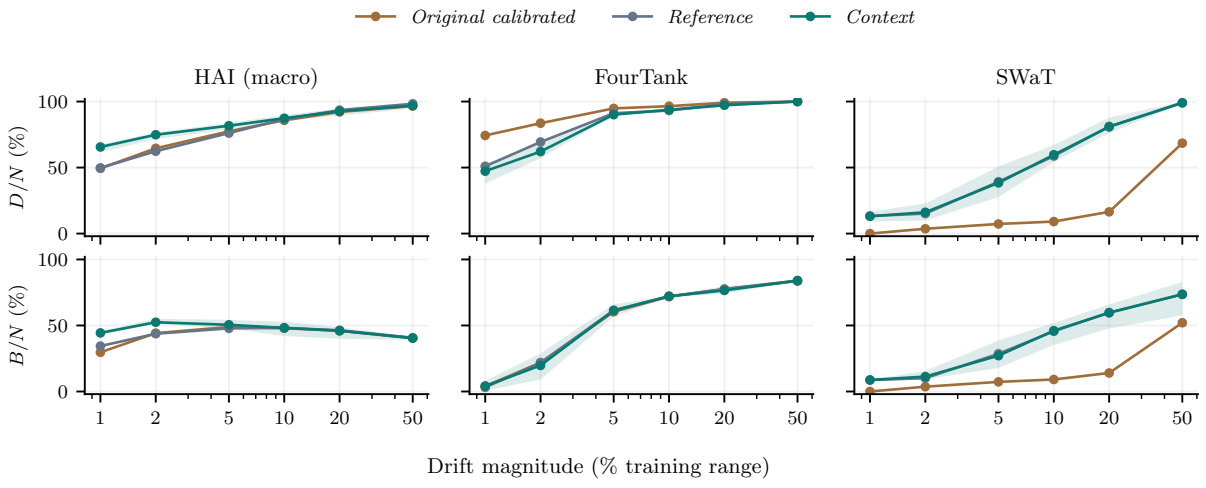


Figure S1: The panels show detection D/N (top) and correct-beneficial coverage B/N (bottom) across drift magnitudes. The rates pool both signs and all checkpoints. For HAI, configurations receive equal weight. The shading spans the five *Context* checkpoint rates.

E3. Temporal assessment and normal damage

The temporal panel contains all 712 planned observations per policy, comprising 616 faults and 96 unchanged controls across checkpoints 23 and 42. The observations come from separate time intervals of the same SWaT normal recording used in the primary study. This within-recording comparison retains its own denominators.

The two steady fault blocks yield 44 beneficial proposals under both *Context* and *Reference* out of 308 faults. In the two transition/settling blocks, the corresponding counts are 33 and 21 out of 308. The beneficial gain occurs at both checkpoints, but all eight harmful fault proposals occur at checkpoint 23 under both policies. The additional signed benefit is concentrated in two source/target stuck-fault patterns at checkpoint 42. Each pattern is repeated as exact, noisy and quantized stuck. Together these repetitions account for about 97.8% of the incremental signed benefit. They are not independent demonstrations of six distinct mechanisms.

Table S4: Normal outcomes in the temporal panel. Every proposal is harmful, and each policy covers all 96 unchanged observations. The mean signed benefit averages over all 96 controls and is expressed in percent of training range.

Policy	Proposals	Harmful	Mean signed benefit (%)
Context	2	2	−1.48
Reference	1	1	−0.85
Residual dominance	0	0	0.00

The shared harmful normal proposal selects channel 1 in the block starting at source row 439405 and has after-error 0.814176413 of training range. The additional *Context* proposal selects channel 2 in the block starting at row 434905 and has after-error 0.605110522 of training range. Both occur at checkpoint 42. The before-errors are zero because these observations are unchanged controls. The additional error cannot be dismissed solely because it arises from one proposal.

E4. Control families, concern and graph limits

The complete control panel contains 1,335 observations per policy. The table preserves the distinctions between unchanged inputs, witness-channel corruption, common-mode corruption, invalid context and the three native qualification categories.

Table S5: The complete *Context* control matrix. Here D counts numeric events, C retained concern, N_P proposals and N_V process-vetoed previews. The columns Q_P and Q_V count estimates with available clean-reference quality. A zero in a native quality column means no clean-reference quality is available, rather than measured reconstruction failure or success.

Control	N	D	C	N_P	N_V	Q_P	Q_V
Unchanged	660	102	161	47	5	47	5
Common mode	120	105	105	55	0	55	0
Witness corruption	120	105	105	78	0	78	0
Invalid context	180	135	180	0	0	0	0
Native supported	25	24	25	15	0	0	0
Native unverified	225	154	163	53	49	0	0
Native no response	5	0	0	0	0	0	0

Retaining the native AE alarm increases concern on unchanged observations from 109 to 161 out of 660, while the 47 proposals remain unchanged. All 47 normal proposals and five normal previews worsen the selected channel. These counts refer to observations rather than grouped alarms or measured operator workload. The primary preview branch exposes 40,696 candidates withheld by the process veto. Of these, 37,779 are correct-beneficial and 2,917 harmful. The correct-beneficial fraction is 92.83%.

The witness controls produce 78 proposals, of which 74 improve the selected channel and four degrade it. The common-mode controls produce 55 proposals, of which 54 improve the selected channel and one degrades it. The second corrupted common-mode channel is not assessed. These results therefore establish selected-channel outcomes, not complete common-mode recovery or process protection.

The response-supported native subset contains 25 checkpoint observations from five source windows. All retain concern, 15 receive proposals and none triggers the process flag F. The absence of clean-native counterfactuals prevents a quality judgement on these estimates. The panel provides no positive demonstration of process-veto utility.

The context-validity check U concerns availability, alignment and validity under an assumed trusted source. A low-support warning or unseen-state pooled normalization is distinct from invalid context. The graph-dependent flag can remain zero when distributed persistent peaks lack a connected declared ownership graph. For the retained native example `26eefd15ed54d0b6cffb`, all thirteen channels show persistent peaks, but the graph is disconnected or incomplete and $F = 0$. Detailed graph-status reasons are available for that one illustrative observation. The retained native prediction cache does not store those reasons for the other 254 native checkpoint observations, so their detailed statuses remain unavailable.

M1. Matched decision and illustrative trace identities

The useful example is FourTank case `292fd580e6f4676f9bfd`, checkpoint 23. The selected channel is LIC23002, with retained channel index 6. The controller changes at row 22 of the 300-row observation and subsequently remains in state 8 throughout the five scoring windows. The history still records the time spent in that state. The policies share the AE, scaler, signed residuals, candidate estimates, ranking, valid-candidate status and required gap. The context is valid and the process flag is zero under both policies.

Table S6: The matched useful observation. The last-window signed mean m concerns LIC23002. The event/proposal row gives A and P separately for each policy. Only the final row uses paired clean values to evaluate the completed decisions.

Quantity	*Reference*	*Context*
Last-window signed mean m	2.19	3.63
Event score S	21.47	59.08
Calibrated event limit	24.73	36.00
Candidate gap Δ	0.97	0.97
Required gap τ	0.80	0.80
Event A / proposal P	0 / 0	1 / 1
Post-policy MAE (% range)	1.00	0.43

The candidate gap is approximately 0.9690840738 against the common requirement of 0.8. The last-window *Reference* centre and scale are −0.0001140481 and 0.0039951881. The corresponding *Context* values are −0.0013751177 and 0.0027534988. The more negative expected offset and smaller normal scale make the observed positive residual more pronounced after standardization. The *Context* score crosses its own higher event limit while *Reference* remains below its limit. The changed event admission permits the same candidate estimate to be proposed. The evaluator confirms that its MAE falls from 1.00% to 0.43% of training range.

The adverse example is FourTank case `0161acc7b63f0df69ca0`, also at checkpoint 23. Its injected drift affects channel 0, while the proposal selects channel 7 and adds 76.17% error to that clean channel. Both examples pass the context and candidate checks without triggering the process flag. They illustrate the mechanism and its failure, not the prevalence of either outcome.

The retained selection rule orders cases by lexicographic configuration, then numeric checkpoint seed, then lexicographic case ID. The useful case is the first correct-beneficial primary *Context* proposal for which *Reference* abstains. The adverse case is the first strictly harmful primary *Context* proposal. The rule uses evaluator outcomes and was fixed for the retained illustrations. The accompanying E5 trace catalog retains observable history, centre/scale trajectories, candidate evidence and separate evaluator arrays. The complete window-level figure for both examples is generated as `window_detail.pdf` by `scripts/reproduce_figures.py`. It preserves the residual-energy and signed-accumulation trajectories together with the routing flags and candidate gaps. The local calculation source is `scripts/matched_case.py`. Its documented command accepts an unpacked closed-evidence package and writes reporting output outside that package.